

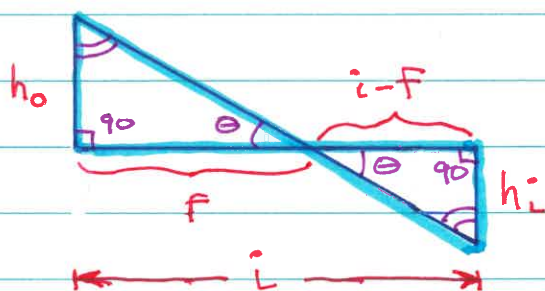
Lecture 5

1/30/26
JRA

Homework Hints

Discussed two homework problems from homework assignment due today

- Problem 3.7 Derivation of Thin Lens Equation



$$\tan \theta = \frac{h_o}{f} \quad \text{Upper triangle}$$

$$\tan \theta = \frac{h_i}{i-f} \quad \text{Lower triangle}$$

$$\frac{h_o}{f} = \frac{h_i}{i-f} \Rightarrow \boxed{\frac{h_o}{h_i} = \frac{f}{i-f}}$$

Repeat process for yellow triangles in book

This problem can be solved using similar triangles OR the definition of tangent

- Problem 3.1b Two-Lens System

(a) Note: f is negative for a diverging lens

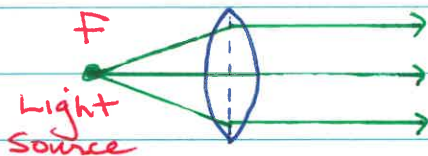
(b) When image from 1st lens is object for 2nd lens, compute distance to 2nd

Here, for diverging object is $48 - 30 = 18$ cm away

(c) Look at ray tracing tutorial & sign conventions

Homogeneous Illumination

To obtain homogeneous light, put light source at focal point



Can use same diagram going either way.
 Confirm w/ Snell's Law.

See Demo in last lecture.

Image Quality

Image quality is a pervasive issue w/ all imaging techniques, so we'll revisit it again & again.

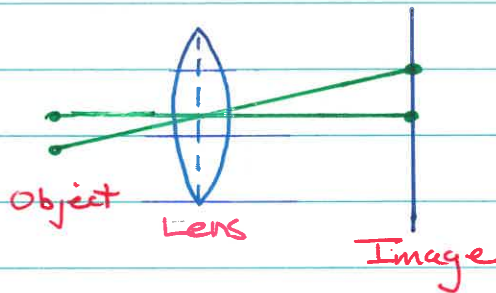
One key measure of quality is "resolution."

The determinants of resolution vary w/ technique, but diffraction appears over & over as a major determinant.

What is resolution? Subtle.

It's standard in microscopy to use a one-parameter descriptor of resolution.

Ray optics - NO DIFFRACTION (because it's ignored)
 $\Rightarrow$ NO resolution problem



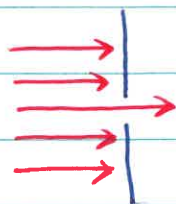
Two points in object map into two points in image, in the absence of diffraction

Resolution - can you tell that two points in the object are still two points when you look at the image?

Ray optics is an approximation: need wave optics to understand resolution. This will come up w/ ultrasound too.

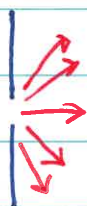
Diffraction is wave optics.

Light & sound are waves. Can see & hear around corners, apertures, etc. because of diffraction. This applies to openings & obstacles.



NO!

Light spreads at hole, so image downstream is NOT just a bright spot.



Little aperture
 $\Rightarrow$ MORE spreading



Big aperture
 $\Rightarrow$ LESS spreading

Key Facts about spreading

- (1) Spreading increases as aperture size decreases
- (2) Spreading increases as wavelength increases
 λ

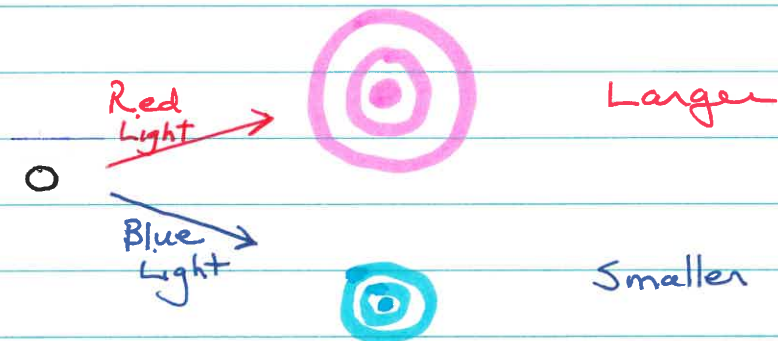
Red light $\Rightarrow$ worse
 Blue light $\Rightarrow$ better

ANIMATION

PHET

Diffraction

Used simulation to examine circular apertures because they correspond to most biological situations



Aperture Size - looked at how aperture size affects the size of the diffraction pattern

Smaller aperture $\Rightarrow$ Larger pattern

Larger aperture $\Rightarrow$ Smaller pattern

Wavelength - looked at how wavelength of light affects the size of the diffraction pattern

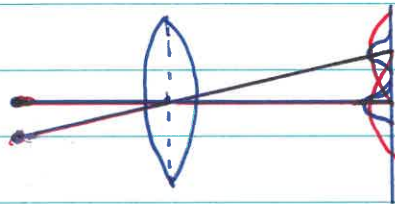
Longer wavelength (red) $\Rightarrow$ Larger pattern

Shorter wavelength (blue) $\Rightarrow$ Smaller pattern

Looked especially at size of central maximum, which comprises about 80% of the total intensity

Geometrical (ray) optics predicts that a point will be imaged as a point

Wave optics predicts spreading due to diffraction $\Rightarrow$ a point is NOT imaged as a point



Blue $\rightarrow$ narrow spots

Red $\rightarrow$ wide spots

Here, blue light patterns are narrow relative to spacing $\Rightarrow$ resolvable despite spacing

Red light patterns are wide & overlap significantly $\Rightarrow$ NOT resolvable due to blurring

Big qualitative take-home messages:

- (1) Want narrow diffraction patterns so closely spaced objects can be resolved
- (2) Resolution improves if aperture (e.g., lens) is large $\Rightarrow$ use big lenses AND if wavelength is short $\Rightarrow$ use blue light

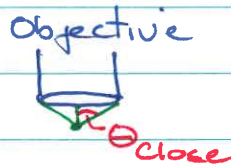
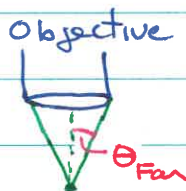
Use $\leftarrow$ Big lenses
Short wavelengths

Want to gather as much light as possible

ANIMATION

Mol Expressions

Numerical Aperture



$$\theta_{\text{close}} > \theta_{\text{Far}}$$

⇒ Closer lens captures a larger cone of light

This is why you usually work with microscope objective very close to specimen

"Numerical Aperture" can be used to quantify the amount of light capture

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Mol Expressions

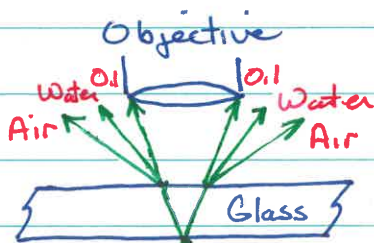
NA $\hat{=}$ Resolution

Higher NA $\Rightarrow$ better resolution (narrower diffraction pattern)

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Mol Expressions

Immersion Oil $\hat{=}$ Refractive Index



Rays bend away from the objective when exiting the glass if the medium between the glass & objective consists of air or water $\Rightarrow$ NOT captured

Rays do NOT bend when oil is the intervening medium

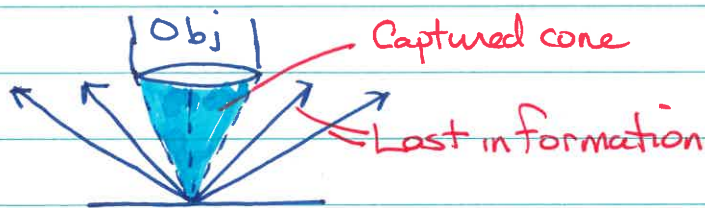
Better capture: Oil > Water > Air

Specimens emit light.

The more light you capture the more information you get & the less distortion the image suffers

⇒ Gather as much light as possible from the specimen

Do NOT want to throw light away.



Want captured cone to have a large capture angle

- Separation - specimen should be close to lens

- Intervening Medium

Oil or water between specimen & lens is helpful.
↑ Best ↑ Good

Oil & water help because they reduce light lost to refraction (bending)

- Wavelength - shorter λ s are better
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